

IE2021 – Requirements Engineering (2026)

Assignment 2: Requirements Specification

Due Date: 17th of October, Friday

Marks: This assignment counts for 10% of the final marks.

Important note

This assignment is to be carried out with the same a group (5 members) and the project topic (ISP) formed for the Assignment 1.

Objective

Assignments' 1 and 2 are closely related to each other. The two assignments together form a complete software requirements specification (SRS). Assignment 2 expects you to further use advanced modelling techniques and then produce a software requirements specification (SRS) document.

Doing the Assignment

1. Refine all functional and non-functional requirements for the new system
2. Develop models for the key aspects of the problem using any free / open-source modelling tools of your choice like Star UML, ArgoUML, PlantUML, OpenOME etc. Your models must include:
 - Structural models
 - i. UML Class Diagram including all classes and relationships
 - ii. UML Physical diagram
 - Behavioural models
 - i. UML Sequence Diagram for 4 selected use cases
 - ii. UML State Diagram for 1 selected object
 - Flow models
 - i. UML Activity Diagrams for 4 selected use cases
 - ii. BPMN Process Models for the above 4 activity diagrams
 - Decision models
 - i. 2 decision tables for selected functions
 - ii. 2 decision trees for the same 2 functions
3. Create wireframe/ mock-up diagrams that illustrate what the requirement or functionality may look like when fulfilled. These must be done using appropriate diagram modelling tools such as draw.io, caco.com or any other suitable tool.

Written requirements:

You should submit ONE report in .pdf format and contain all relevant graphics of your work.

Item	Details
Cover Page	Module Name, Project Name, Group details, Submission date
Introduction	1 page description about the project domain, organization, existing problems etc.

Methodology section	Describe the process of requirements identification, elicitation, and documentation
Discussion section	Briefly describe any difficulties you encountered, limitations on your analysis, interesting lessons learned during the analysis and specification process, etc
Refined List of FR, NFR, Constraints	Using correct template
Models	UML Class Diagram UML Physical Diagram, UML Sequence Diagrams (4) UML Activity Diagrams (4) UML State Diagram Process Models (4) Decision Tables, Trees (4) Provide enough commentary on the models for the reader to understand them.
Wireframes	Mock-up diagrams.
References List	Provide all references to all resources and tools used
Group members contribution table	1 page table indicating which members developed each item
Appendices	

Notes:

- Drawings must be clear and legible. You can use open-source or free modelling tools of your choice like ArgoUML, Star UML, PlantUML etc to draw your models
- Clearly list any assumptions you might make in completing this assignment at appropriate places.
- Make sure you provide enough commentary (and annotations) with the models for the reader to understand what each model is of, and how to understand them.
- Provide any further appendices you feel are relevant.
- Text font size should be 11 points.
- Plagiarism will be penalized.